

SAMBP Technical Report TR-81/2028

Geomagnetic Disturbance and GIC Protection

Lehtinen-Pirjola GIC Flow Model, IEC 60076-7 Transformer Thermal Model, 5-Bus 400 kV
Network: 11 Tests PASS

SAMBP Research Group

Smart Adaptive Multi-Bus Protection Research, IIT Madras

April 2028 | Chapter 3 (Protection Challenges) | Prerequisite: TR-67 (HIL)

Contents

1	Background and Objectives	1
1.1	Motivation	1
1.2	Objectives	1
2	Theory and Methodology	1
2.1	Lehtinen-Pirjola GIC Flow Model	1
2.2	Transformer Thermal Model	2
2.3	Alarm Levels	2
3	Simulation Results	2
3.1	5-Bus 400 kV Network: GIC vs E-Field	2
3.2	11-Test Pass Summary	3
4	Conclusion	3

1 Background and Objectives

1.1 Motivation

Geomagnetic Disturbances (GMD) caused by solar storms drive quasi-DC Geomagnetically Induced Currents (GIC) through transmission networks. GIC saturates power transformers, causing half-cycle saturation, increased reactive power demand, harmonic injection, and potentially catastrophic thermal damage [1]. The Quebec 1989 blackout (6 hours, 9 GW loss) and the Hydro-Quebec transformer damage are canonical examples.

The SAMBP framework adds GMD/GIC protection as a special protection scheme (SPS) separate from the conventional fault-protection functions.

1.2 Objectives

1. Implement the Lehtinen-Pirjola (LP) GIC flow model.
2. Implement the IEC 60076-7:2018 transformer thermal model with GIC.
3. Define alarm levels: NORMAL, ELEVATED, CRITICAL, TRIP.

4. Validate on a 5-bus 400 kV network with 11 unit tests.

2 Theory and Methodology

2.1 Lehtinen-Pirjola GIC Flow Model

The LP model [2] treats the transmission network as a resistive circuit driven by the geoelectric field $\mathbf{E}$ (induced by the time-varying geomagnetic field). The network equations are:

$$(\mathbf{Y}_n + \mathbf{Y}_e) \mathbf{V} = \mathbf{J}_e \quad (1)$$

where $\mathbf{Y}_n$ is the network admittance matrix (DC resistance of lines and transformer windings), $\mathbf{Y}_e$ is the earth admittance matrix (earthing resistances), $\mathbf{V}$ is the vector of node voltages, and $\mathbf{J}_e = \mathbf{Y}_e^{(ij)} \cdot \int_{ij} \mathbf{E} dl$ is the geoelectric source current vector (line integral of $\mathbf{E}$ along each transmission line).

The neutral GIC at bus k is: $I_{\text{GIC},k} = Y_{e,kk} V_k$ (quasi-DC neutral current through transformer).

3-bus LP benchmark. With $R_{\text{line}} = R_{\text{earth}} = 2 \Omega$ and $E_x = 1 \text{ V/km}$: Bus 1: $I_{\text{GIC}} = 25.0 \text{ A}$; Bus 2: 25.0 A ; Bus 3: -50.0 A . KCL check: $\sum I_{\text{GIC}} = 0.000 \text{ A}$. ✓

5-bus 400 kV scaling. GIC scales linearly with E-field intensity: $I_{\text{GIC,max}} = 57.7 \text{ A}$ at $E = 1 \text{ V/km}$; $= 577 \text{ A}$ at $E = 10 \text{ V/km}$ (extreme solar storm).

2.2 Transformer Thermal Model

The IEC 60076-7:2018 thermal model (Thermal Model 1) [3]:

$$\tau_{\text{TO}} \frac{d\Delta\Theta_{\text{TO}}}{dt} = \Delta\Theta_{\text{TO,R}} \left[\frac{K_{\text{eff}}^2 + R}{1 + R} \right]^{n_{\text{TO}}} - \Delta\Theta_{\text{TO}} \quad (2)$$

$$\tau_{\text{HS}} \frac{d\Delta\Theta_{\text{HS}}}{dt} = \Delta\Theta_{\text{HS,R}} K_{\text{eff}}^{2n_{\text{HS}}} - \Delta\Theta_{\text{HS}} \quad (3)$$

where K_{eff} is the effective load factor incorporating GIC:

$$K_{\text{eff}} = \sqrt{K^2 + \left(\frac{I_{\text{GIC}}}{I_{\text{rated}}} \times k_{\text{gic}} \right)^2} \quad (4)$$

k_{gic} is the GIC-to-thermal-equivalent factor (typically 0.5–1.0 depending on transformer design).

2.3 Alarm Levels

Table 1: GIC protection alarm levels based on hot-spot temperature.

Level	Hot-spot Θ_{HS}	Action
NORMAL	$< 98^\circ\text{C}$	None
ELEVATED	$98\text{--}120^\circ\text{C}$	Operator alert, reduce load
CRITICAL	$120\text{--}140^\circ\text{C}$	Automatic load shedding
TRIP	$\geq 140^\circ\text{C}$	Transformer isolation

3 Simulation Results

3.1 5-Bus 400 kV Network: GIC vs E-Field

Table 2: GIC levels and alarm status vs. E-field intensity (5-bus 400 kV, $K = 1$, $\Theta_{\text{amb}} = 25^\circ\text{C}$).

E (V/km)	Max I_{GIC} (A)	Θ_{HS} ($^\circ\text{C}$)	Alarm	Status
0.5	28.9	91.0	NORMAL	✓
1.0	57.7	91.3	NORMAL	✓
2.0	115.5	92.3	NORMAL	✓
5.0	288.7	107.8	ELEVATED	✓
10.0	577.3	127.0	CRITICAL	✓

3.2 11-Test Pass Summary

Table 3: TR-81 unit test results.

Test	Description	Status
T01	LP model: 3-bus KCL check (E_x component)	✓
T02	LP model: 3-bus KCL check (E_y component)	✓
T03	LP model: linear E-field scaling	✓
T04	LP model: 5-bus 400 kV validation	✓
T05	Thermal model: steady-state (no GIC)	✓
T06	Thermal model: GIC-augmented K_{eff}	✓
T07	Alarm: NORMAL level ($I_{\text{GIC}} = 100$ A)	✓
T08	Alarm: ELEVATED level ($I_{\text{GIC}} = 350$ A)	✓
T09	Alarm: CRITICAL level ($I_{\text{GIC}} = 500$ A)	✓
T10	End-to-end pipeline: LP GIC $\rightarrow$ thermal model	✓
T11	Euler integration accuracy ($< 5^\circ\text{C}$ vs analytical)	✓
Total	—	11/11 PASS

Finding 1 (GIC thermal risk begins at $E > 5$ V/km). *For the 5-bus 400 kV network, the transformer hot-spot temperature reaches the ELEVATED alarm threshold at $E = 5$ V/km ($I_{\text{GIC}} = 288.7$ A) and CRITICAL at 10 V/km. The 1989 Quebec event produced estimated $E \approx 10$ V/km at some locations, consistent with the observed transformer damage.*

4 Conclusion

Finding 2 (LP GIC model: exact KCL verified). *The Lehtinen-Pirjola GIC solver satisfies KCL ($\sum I_{\text{GIC}} = 0$) exactly for all tested E-field configurations, confirming numerical correctness. GIC scales linearly with E-field intensity as expected from the linear LP equations.*

Finding 3 (IEC 60076-7 thermal model: 11 tests PASS). *The IEC 60076-7:2018 thermal model with GIC-augmented load factor correctly tracks hot-spot temperature with Euler integration error $< 5^\circ\text{C}$ vs. the analytical solution.*

References

- [1] C. T. Gaunt and G. Coetzee, “Transformer failures in regions incorrectly considered to have low GIC-risk,” in *Proc. IEEE PowerTech*, Lausanne, 2007.

- [2] M. Lehtinen and R. Pirjola, “Currents produced in earthed conductor networks by geomagnetically-induced electric fields,” *Ann. Geophys.*, vol. 3, no. 4, pp. 479–484, 1985.
- [3] IEC 60076-7:2018, *Power Transformers — Part 7: Loading Guide for Mineral-Oil-Immersed Power Transformers*, IEC, Geneva, 2018.
- [4] NERC, *Special Reliability Assessment: Effects of Geomagnetic Disturbances on the Bulk Power System*, Atlanta, GA, 2012.
- [5] D. H. Boteler and R. J. Pirjola, “Modelling geomagnetically induced currents,” *Space Weather*, vol. 15, no. 1, pp. 258–276, 2017.